

Harshita Karnam

Engineering Undergraduate | Full-Stack Developer | Reinforcement Learning Researcher

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Portfolio: portfolio.vercel.app

PROFESSIONAL SUMMARY

Second-year B.Tech CSE student at IIIT Sri City with interests in AI and intelligent traffic systems. Experienced in full-stack web development, building and deploying production-grade applications, and conducting research at the intersection of reinforcement learning, graph neural networks, and large language models. Published research (under review) on LLM-guided RL for adaptive traffic control. Known for strong problem-solving instincts, effective teamwork, and clear communication of complex technical concepts.

EDUCATION

B.Tech — Computer Science & Engineering *2024 – 2028 (Completed 2nd Year)*
Indian Institute of Information Technology, Sri City, India

TECHNICAL SKILLS

Languages Python, C, C++, HTML, CSS, JavaScript
ML / RL PyTorch, OpenAI Gym, SUMO
Algorithms DQN, Double DQN, PPO, CoLight, Multi-Agent RL
Concepts Graph Attention Networks, LLM Integration, Reward Shaping, Policy Gradient
Tools Git / GitHub, Linux, VS Code

RESEARCH

SafeGAT (*Manuscript Under Review*) — LLM-augmented Graph Attention RL framework for adaptive traffic signal control; outperforms pure RL baselines on unseen road topologies.

PROJECTS

SafeGAT — LLM-Integrated RL Traffic Signal Control *PyTorch, SUMO, LLM*

- Hybrid architecture coupling LLMs with GAT-based RL for adaptive traffic control; improved generalisation across unseen intersections via LLM-derived auxiliary policy guidance.
- Manuscript under peer review; second-year undergraduate work.
- GitHub: [harshitak13/SafeGat](https://github.com/harshitak13/SafeGat)

RL-Based Traffic Signal Control *DQN / DDQN / PPO / CoLight*

- Benchmarked four RL algorithms across single and 4×4 multi-intersection grids; DDQN and CoLight led on reward, queue length, and travel time.
- GitHub: [harshitak13/RL-TSC](https://github.com/harshitak13/RL-TSC)

DQN CartPole — Deep Q-Network from Scratch *PyTorch, OpenAI Gym*

- Built a full DQN agent with experience replay and target network on CartPole-v1; ablation study vs. vanilla Q-learning validated replay buffer and target-network stabilisation.
- GitHub: [harshitak13/DQN_Cartpole](https://github.com/harshitak13/DQN_Cartpole)

EXPERIENCE

Director, Management Department — Entrepreneurship Cell, IIIT Sri City *Sept 2025 – May 2026*
Organised AI/ML workshops and tech events; led cross-functional student teams.

Member, Developer Relations — Google Developer Groups (GDG) *Sept 2025 – May 2026*
Participated in developer events, study jams, and collaborative ML sessions.